A comment on Hong, Chua, and Freeman (2024), "*Reducing Facial Stereotype Bias in Consequential Social Judgments: Intervention Success With White Male Faces*"

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Abstract

The purpose of this replication is to check the robustness of findings of the study "reducing facial stereotype bias in consequential social judgments" conducted by Hong, Chua, & Freeman (2024) on adult participants recruited from MTurk and Prolific, targeted to test whether real-world biases can be mitigated through counterstereotype training in four experimental studies. These four studies are about (a) the effectiveness of counterstereotype training in diminishing the relationship between perceived trustworthiness and courtroom sentencing (life in prison or death), (b) prediction of the training reduce participants' biases in hypothetical sentencing judgments, (c) testing whether the training is successful in implicit trustworthiness evaluations, and (d) extending the training to real-world-like scenarios to determine the guilt of a defendant. The results from studies 1A.1, 1B1, and 2A reported by the authors match with our own reproductions (except for very minor rounding differences), but we were unable to replicate results from 1A2, 1B2, and 2B due to a lack of appropriate code. Based on this replication, it can be concluded that although the values differ slightly from the original study results, where code is available, they do not substantially change the authors' conclusions. We also conducted a robustness reproducibility check by applying a non-standardized asinh transformation to some variables the standardized rating of trustworthiness, and these transformations did not substantially change the authors' conclusions.

1. Introduction

The purpose of this replication was to assess the robustness of the findings in the paper “reducing facial stereotype bias in consequential social judgments" conducted by Hong, Chua, & Freeman (2024) on adult participants that underwent a counterstereotype training to reduce their reliance on facial appearance in consequential social judgments of white male faces. Past studies either attempted to provide cues to participants that are more useful than facial appearance or they have explicitly educated participants to stop using facial appearance—an approach that has proved unsuccessful. Recent studies using interventions operating on participants’ underlying associations of facial appearances with overt manifestations of trustworthiness or untrustworthiness attributes in a more automatic, implicit manner (i.e., using a counterstereotype paradigm), have shown promise that this study aims to use as an intervention.

The researchers attempted to test whether these real-world biases can be mitigated through counterstereotype training using randomly selected samples under four online experiments.

Study 1 finds that the training is effective in diminishing the relationship between perceived trustworthiness and courtroom sentencing—that is, whether prison inmates are sentenced to life in prison or death— and this bias reduction cannot be attributed to merely disengaging one’s attention from people’s faces.

Study 2 extends the findings to sentencing-severity judgments, predicting that the training will reduce participants’ biases in hypothetical sentencing judgments.

Study 3 uses a sequential priming paradigm to test whether such training is successful also in automatic, implicit trustworthiness evaluations.

Study 4 extends the training to real-world-like scenarios in which participants determine the guilt of a defendant in the presence of realistic decision-relevant information.

The authors analyzed the data using mixed-effects binomial regressions with interactions. The paper estimated the value of consequential social judgments for the target groups to predict the real-world sentencing outcomes and the main results convey beta coefficients, standard errors, *p* values, and 95% confidence intervals in the four studies. Our replication computed these estimated values using their data set and checked their consistency and robustness.

This replication proceeds as follows. In section 2 we assess the computational reproducibility of the results conveyed in the paper’s studies 1 and 2. In section 3 we perform robustness checks for the main results of study 1.

2. Computational Reproducibility

Instructions:

We used the replication package available online: https://osf.io/6m9jv/. The analysis code and data were both complete, and the scripts were in the form of rmarkdown documents.

The paper reported no tables of results, but we successfully computationally reproduced all main results from the text of the paper. However, we observed the following errors when running the code, which had to be corrected:

Line of code (file)	Error	Correction	Explanation
93 (Study 1_osf.Rmd) Study 1	> t.test(rt ~ train,df1b_n,var.equal=T) Error in eval(predvars, data, env) : object 'train' not found	t.test(rating ~ Group,df1b_n,var.equal=T)	Wrong variable names
163 (Study 1_osf.Rmd) Study 1 Same for Study 2 (line 84), Study 3 (line 227), and Study 4 (line 89)	> png("~/Desktop/figures2.png", width = 20, height = 10, units = 'in', res = 300) Error in png("~/Desktop/figures2.png", width = 20, height = 10, units = "in", : unable to start png() device	png("figures2.png", width = 20, height = 10, units = 'in', res = 300)	Wrong directory for saving figure

Original results

Study 1A

Study 1A.1

The paper does not report any tables of results, so we report our own table from the script matching the text results:

Table 1

Parameter	*b*	*z*	*p*	OR	95% CI
tr z	-0.04	-5.27	< .001***	0.96	[0.95, 0.98]
train z	-0.01	-0.39	.694	0.99	[0.97, 1.02]
tr z × train z	0.05	3.88	< .001***	1.06	[1.03, 1.09]

Note. tr = trustworthiness; train = training

Original Text: The main effect of trustworthiness was significant, b = −0.04, SE = .01, z = 5.27, p < .001, OR = 0.96, 95% confidence interval (CI) = [.95, .98], indicating that faces that were rated less trustworthy were more likely to belong to individuals who were sentenced to death than life in prison. The main effect of training was not significant, b = −0.01, SE = .01, z = 0.39, p = .69, OR = 1.00, 95% CI = [.97, 1.02]. Critically, there was a significant interaction, b = 0.05, SE = .01, z = 3.87, p < .001, OR = 1.06, 95% CI = [1.03, 1.09].

Conclusion: the results match (except for very minor rounding differences).

Study 1A.2

There is no code to reproduce the results below.

Original Text: The effect of perceived trustworthiness was highly significant among control participants, simple b = −0.06, SE = .01, z = 6.62, p < .001, OR = 0.94, 95% CI = [.92, .96], indicating that faces that were rated less trustworthy were more likely to belong to individuals who were sentenced to death than life in prison (Fig. 2a). However, in the trained condition, trustworthiness was no longer predictive of real-world sentencing outcomes, simple b = −0.01, SE = .01, z = 0.96, p = .34, OR = 0.99, 95% CI = [.97, 1.01].

Conclusion: We cannot replicate those results from the available code.

Study 1B

Study 1B.1

Self-generated table:

Table 2

Parameter	*b*	*z*	*p*	OR	95% CI
att z	-0.03	-3.81	< .001***	0.97	[0.95, 0.98]
train z	0.01	0.73	.463	1.01	[0.98, 1.04]
att z × train z	0.01	0.67	.505	1.01	[0.98, 1.05]

Note. tr = trustworthiness; train = training

Original Text: The main effect of attractiveness was significant, b = −0.03, SE = .01, z = 3.81, p < .001, OR = 0.97, 95% CI = [.95, .98]; faces that were rated less attractive were more likely to belong to individuals who were sentenced to death than life in prison. The main effect of training was not significant, b = 0.01, SE = .01, z = 0.73, p = .47, OR = 1.01, 95% CI = [.98, 1.04]. Critically, the interaction was also not significant, b = 0.01, SE = .02, z = 0.70, p = .49, OR = 1.01, 95% CI = [.98, 1.05].

Conclusion: the results hold for the most part, except for some minor rounding differences detailed in the table below:

Table 3

Parameter	**Original study**		**Conceptual Reproduction**	
	z	p	z	p
train z	0.73	.47	0.73	0.46
att z × train z	0.70	.49	0.67	.505

Differences are in the p-values for the training z variable (.1) as well as the interaction effect (0.015). In addition, the estimate for the interaction differs by .03. These minor differences could be attributed to changes in the packages used to perform the estimation, or to slight changes in the script or data used and what is actually reported in the paper. However, we note that we ran the statcheck::checkPDF() function on the paper, and statcheck found no statistical mistake in the available data.

Study 1B.2

There is no code to reproduce the results below.

Original Text: The effect of perceived attractiveness on sentencing outcome did not differ between control participants, simple b = −0.04, SE = .01, z = 3.10, p = .002, OR = 0.96, 95% CI = [.94, .99], and trained participants, simple b = −0.03, SE = .01, z = 2.25, p = .02, OR = 0.97, 95% CI = [.95, 1.00]; see Figure 2b.

Conclusion: We cannot replicate those results from the available code.

Study 1B.3

Original Text: Collapsing the data of Studies 1a and 1b and testing for a three-way interaction confirmed that **the lack of a two-way interaction in attractiveness ratings of Study 1b was statistically different from the significant two-way interaction in trustworthiness ratings of Study 1a** (see Supplemental Analysis S2 in the Supplemental Material).

Conclusion: the text above does not provide any statistical value, only a statement that the two interactions were not different from each other. We can only then reproduce this statement by verifying that the *p-value* of the difference is not significant.

Note: Also, the link between this statement and the code provided in S2 is not clear.

When comparing supplemental material differences in interactions S1 and S2 through an ANOVA, the Chi-Square statistic is 0, with 4 degrees of freedom, and a p-value of 1. This indicates that there is no significant difference between the models. In other words,

the additional parameters in mod3a do not provide a significantly better fit to the data compared to mod3b.

Study 2

Study 2A

Self-generated table:

Table 4

Parameter	*B*	*z*	*p*	OR	95% CI
sv z	0.10	10.31	< .001***	1.10	[1.08, 1.13]
train z	0.004	0.30	.766	1.00	[0.98, 1.03]
sv z × train z	-0.05	-2.75	.006	0.95	[0.92, 0.99]

Note. sv = sentencing severity; train = training

Original text: The main effect of sentencing recommendation was significant, b = 0.10, SE = .01, z = 10.35, p < .001, OR = 1.10, 95% CI = [1.08, 1.13]. The main effect of training was not significant, b = 0.004, SE = .01, z = 0.30, p = .77, OR = 1.00, 95% CI = [.98, 1.03]. Critically, the predicted interaction was significant, b = −0.05, SE = .02, z = 2.75, p = .006, OR = 0.95, 95% CI = [.92, .99].

Conclusion: the results hold for the most part, except for some minor differences in the z-score for the sentencing severity estimate. In our results, the value is 10.31, whereas the authors obtained 10.35.

Study 2B

There is no code to reproduce the results below.

Although control participants' sentencing recommendations were strongly predictive of whether inmates were actually sentenced to death versus life in prison, simple b = 0.12,

SE = .01, z = 9.22, p < .001, OR = 1.13, 95% CI = [1.10, 1.16], this relationship was substantially attenuated among trained participants, simple b = 0.07, SE = .01, z = 5.51, p < .001, OR = 1.08, 95% CI = [1.05, 1.10]

Conclusion: We cannot replicate those results from the available code.

3. Robustness Reproducibility

Discrepancies Between Pre-analysis Plan and Article

The authors did not register a pre-analysis plan.

Robustness check 1: Applied Transformation

Study 1A

For our first robustness reproducibility check, we applied a transformation to the only non binary variable, the standardized rating of trustworthiness, an ordinal variable with seven possible ratings. This choice of sensitivity analysis was decided after looking at the data (as the original plan was to impute missing data or check for outliers, but there were no missing data or outliers). We selected the transformation to apply automatically via the BestNormalize R package algorithm, which chose a non-standardized asinh(x) transformation. We report the result below for Study1A.1:

Table 5 (Table 1 with a non-standardized asinh(x) transformation on tr)

Parameter	*b*	*z*	*p*	OR	95% CI
tr z	-0.04	-5.18	< .001***	0.96	[0.94, 0.97]
train z	-0.01	-0.36	.720	1.00	[0.97, 1.02]
tr z × train z	0.07	3.99	< .001***	1.07	[1.04, 1.11]

Note. tr = trustworthiness; train = training

Table 1 (Original)

Parameter	*b*	*z*	*p*	OR	95% CI
tr z	-0.04	-5.27	< .001***	0.96	[0.95, 0.98]
train z	-0.01	-0.39	.694	0.99	[0.97, 1.02]
tr z × train z	0.05	3.88	< .001***	1.06	[1.03, 1.09]

Note. tr = trustworthiness; train = training

Conclusion: The only data to which a transformation could be somewhat meaningfully applied, the ordinal data, was already approximately normally distributed, so the transformation did not end up changing the values substantially. Therefore, although the values differ slightly, it does not change the overall conclusions or significance thresholds.

Study 1B

For study 1B, we apply the same robustness reproducibility check, with a transformation to the only non binary variable, the standardized rating of trustworthiness, an ordinal variable with seven possible ratings. We selected the transformation to apply automatically via the BestNormalize R package algorithm, which chose a non-standardized asinh(x) transformation. We report the result below for Study1B.1:

Table x (Table 2 with a non-standardized asinh(x) transformation on tr)

Parameter	*b*	*z*	*p*	OR	95% CI
att z	-0.04	-3.75	< .001***	0.96	[0.94, 0.98]

train z	0.01	0.73	.464	1.01	[0.98, 1.04]
att z × train z	0.01	0.69	.491	1.02	[0.97, 1.06]

Note. att z = trustworthiness; train = training

Table 2 (Original)

Parameter	*b*	*z*	*p*	OR	95% CI
att z	-0.03	-3.81	< .001***	0.97	[0.95, 0.98]
train z	0.01	0.73	.463	1.01	[0.98, 1.04]
att z × train z	0.01	0.67	.505	1.01	[0.98, 1.05]

Note. tr = trustworthiness; train = training

Conclusion: The only data to which a transformation could be somewhat meaningfully applied, the ordinal data, was already approximately normally distributed, so the transformation did not end up changing the values substantially. Therefore, although the values differ slightly, it does not change the overall conclusions or significance thresholds.

Reference

Hong, Y., Chua, K. W., & Freeman, J. B. (2024). Reducing facial stereotype bias in consequential social judgments: Intervention success with White male faces. *Psychological Science*, *35*(1), 21-33.